The 66-value policy observation

every channel is something a real machine could measure

terrain profile — channels 0:9
nine forward height bins ahead of the cutter

task error — channels 9:12
cut-height and lateral error, cutter vertical velocity

cylinder state — channels 12:28
four extensions, four velocities, low/high travel headroom

actuator interface — channels 28:44
previous action, inferred valve, predicted increment, target lead

pose and context — channels 44:54
head alignment and rate, truck speed, scan validity, head axis, contact force, path preview

model error — channels 54:66
cylinder tracking residual, actuator prediction residual, normalized effort